

Microfluidics --- Solution

X23 (2023) — English translation

- A1** Let there be a steady laminar flow of water in the capillary. Find the dependence of the liquid flow velocity $v(z)$ in the capillary on the coordinate z for steady laminar flow of water in the capillary. Axis z is directed along side b , the origin of coordinates is on one of the walls of the capillary. Express your answer in terms of ΔP , capillary parameters and liquid viscosity η . **1.00**

Let us consider a thin layer of liquid enclosed between planes with coordinates $z_1 = z$ and $z_2 = z + dz$. Viscous friction forces acting in the corresponding planes:

$$f_1 = -\eta aL \frac{dv_x}{dz}(z) f_2 = \eta aL \frac{dv_x}{dz}(z + dz)$$

Тогда второй закон Ньютона примет вид:

$$\Delta P a dz + f_1 + f_2 = 0 \quad (1)$$

Отсюда уравнение на скорость:

$$\frac{d^2 v_x}{dz^2} = -\frac{\Delta P}{\eta L} \quad (2)$$

Заметим, что совершенно необязательно было устремлять толщину слоя к нулю (т.е. уравнение (1) верно и при конечном dz). Более того, зафиксировав (например $z_1 = 0$ или $z_1 = b/2$), мы получим проинтегрированный аналог уравнения (2). Общее решение уравнения (2) имеет вид:

$$v_x = -\frac{\Delta P}{\eta L} \frac{z^2}{2} + C_1 z + C_2$$

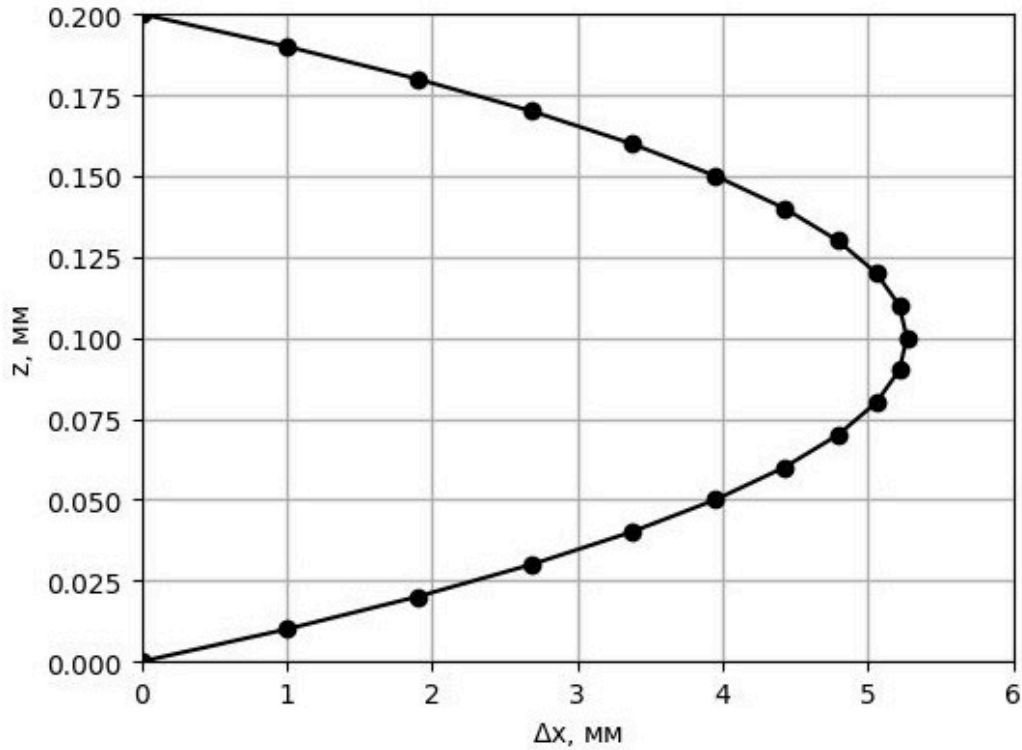
Константы интегрирования определяются граничными условиями, в данном случае $v(0) = 0$ и $v(b) = 0$. From here we get the correct answer

Answer:

$$v(z) = \frac{\Delta P}{2\eta L} (bz - z^2)$$

- A2** At some point, a reagent is uniformly injected into the initial section of the capillary AA' . In the figure, thick horizontal lines indicate the walls of the capillary. On your answer sheet, draw where the reactant will be located via $\tau = 5 c$. For the drawing, select an appropriate scale along the horizontal axis, which is directed along the fluid flow. Capillary parameters: $L = 8 \text{ мм}$, $b = 0.2 \text{ мм}$, pressure difference $\Delta P = 1.5 \text{ Па}$, $\eta = 8.9 \cdot 10^{-4} \text{ Па} \cdot \text{с}$. Neglect the diffusion of the reagent. **1.00**

Let's plot the dependence $x(z) = \tau v(z)$ on the graph



- B1** Find the steady-state velocity of ions $v_z(z)$ at a distance z from the capillary boundary. **0.50**
 Assume that the movement of the ion during the establishment of the velocity is small.
 Express your answer in terms of the charge of the ion q ($q = \pm e$), α and the potential $\varphi(z)$ (and its derivatives).

In steady state, Newton's second law in projection onto the z axis looks like this:

$$-\alpha v_z + qE_z = 0$$

Выражая электрическое поле через градиент потенциала $E_z = -\frac{d\varphi}{dz}$, we get the answer

Answer:

$$v_z = -\frac{q}{\alpha} \frac{d\varphi}{dz} \quad (3)$$

- B2** Express the concentrations of cations (n_+) and anions (n_-) near the capillary wall in **1.50**
 terms of the potential at this point and also in terms of n_0, e, α, k, T .

The flow of particles associated with the presence of an electric field is balanced by the flow of ions associated with diffusion. Thus, the balance condition will look like this:

$$nv_z - D \frac{dn}{dz} = 0$$

Подставляя сюда уравнение (3), получим

$$-\frac{q}{\alpha D} \frac{d\varphi}{dz} = \frac{1}{n} \frac{dn}{dz}$$

Интегрируя обе части уравнения по z и подставляя $D = \frac{kT}{\alpha}$, находим

$$-\frac{q}{kT}(\varphi - 0) = \ln\left(\frac{n}{n_0}\right)$$

As a result,

The result of this paragraph is a special case of the Boltzmann distribution. Of course, this coincidence is not accidental.

Answer:

$$n = n_0 \exp\left(-\frac{q\varphi}{kT}\right)$$

B3 Express the volume density of the uncompensated charge $\rho(z)$ near the capillary wall in 0.50 terms of potential $\varphi(z)$, ion concentration n_0 and physical constants e, k, T . Assuming that the potential applied to the wall is small ($e\varphi_0 \ll kT$), you will obtain an approximate value of $\rho(z)$.

$$\rho = e(n_+ - n_-) = en_0 \left[\exp\left(-\frac{e\varphi}{kT}\right) - \exp\left(\frac{e\varphi}{kT}\right) \right] = -2en_0 \sinh\left(\frac{e\varphi}{kT}\right) \quad (4)$$

Раскладывая выражение по параметру $\frac{e\varphi}{kT}$, we find

Answer:

$$\rho \approx -2 \frac{e^2 n_0 \varphi}{kT} \quad (5)$$

B4 Find the dependence of the potential $\varphi(z)$ on the coordinate near the capillary wall in the diffuse layer. Remember that the potential at the boundary is φ_δ . Express your answer in terms of n_0, φ_δ, T , physical constants and electrolyte parameters. 1.00

Gauss's theorem in differential form in the one-dimensional case has the form:

$$\rho = -\varepsilon\varepsilon_0 \frac{d^2\varphi}{dz^2}$$

Используя разложение (5) получаем

$$\frac{d^2\varphi}{dz^2} = 2 \frac{e^2 n_0}{\varepsilon\varepsilon_0 kT} \varphi \quad (6)$$

Это линейное дифференциальное уравнение, его общее решение в данном случае выглядит следующим образом:

$$\varphi(z) = C_1 \exp(z/\lambda) + C_2 \exp(-z/\lambda),$$

где λ определяется из уравнения (6), а коэффициенты C_1 и C_2 – из начальных условий $\varphi(0) = \varphi_\delta$ и $\varphi(+\infty) = 0$. Заметим, что второе граничное условие автоматически подразумевает $\lambda \ll b$, т.е. нахождение второй стенки на бесконечности. При учёте отклонения от этого условия, строго говоря, $C_1 \neq 0$. However, the answer to point B5 allows us to cast aside any doubts about this. As a result

Примечание. Let us note a more interesting point from the point of view of physics. For expansion (5) and characteristic values of $\varphi_\delta \sim 1 B$ to be applicable, temperature T must be $T \gg \frac{e\varphi_\delta}{k} \sim 10^4 K$! However, the approximation used in the problem is not as bad as it might seem at first glance. Let us verify this by solving the equations without using approximation (5). From combination (4) and Gauss's theorem we find:

$$\frac{d^2\varphi}{dz^2} = 2 \frac{en_0}{\varepsilon\varepsilon_0} \sinh\left(\frac{e\varphi}{kT}\right)$$

Далее, интегрируя уравнение по φ :

$$\int \sinh\left(\frac{e\varphi}{kT}\right) d\varphi = \frac{kT}{e} \cosh\left(\frac{e\varphi}{kT}\right) + C \int \frac{d^2\varphi}{dz^2} d\varphi = \int \frac{d\varphi'_z}{dz} \varphi'_z dz = \int \varphi'_z d\varphi'_z = \frac{(\varphi'_z)^2}{2} + C,$$

где $\varphi'_z \equiv \frac{d\varphi}{dz} = -E_z$. Подставляя граничное условие $\varphi'_z(+\infty) = 0$ получаем:

$$\varphi'_z = -2\sqrt{\frac{en_0}{\varepsilon\varepsilon_0} \frac{kT}{e} \left(\cosh\left(\frac{e\varphi}{kT}\right) - 1\right)} = -2\sqrt{\frac{en_0}{\varepsilon\varepsilon_0} \frac{kT}{e}} \sinh\left(\frac{e\varphi}{2kT}\right)$$

Далее, разделяя переменные и интегрируя:

$$-2\sqrt{\frac{en_0}{\varepsilon\varepsilon_0} \frac{kT}{e}} z = \int_{\varphi_\delta}^{\varphi} \frac{d\varphi}{\sinh\left(\frac{e\varphi}{2kT}\right)} = \frac{2kT}{e} \ln \tanh \frac{e\varphi}{4kT} \Big|_{\varphi_\delta}^{\varphi}$$

Отсюда

$$\varphi = \frac{4kT}{e} \tanh^{-1} \left[\tanh\left(\frac{e\varphi_\delta}{4kT}\right) \exp(-z/\lambda) \right],$$

где λ в точности равна своему значению в предыдущем рассмотрении. Теперь проанализируем асимптотику данного решения на бесконечности. Для $z \gg \lambda$ экспонента оказывается малой, поэтому в ведущем приближении $\tanh^{-1}(x) \approx x$, и реальная зависимость оказывается

$$\varphi \rightarrow \frac{4kT}{e} \tanh\left(\frac{e\varphi_\delta}{4kT}\right) \exp(-z/\lambda)$$

Таким образом, наше приближение на низких температурах даёт правильное значение λ , however the coefficient before the exponent (prefactor) will be inaccurate.

Answer:

$$\varphi(z) = \varphi_\delta \exp(-z/\lambda), \quad (7)$$

$$\text{где } \lambda = \sqrt{\frac{\varepsilon \varepsilon_0 k T}{2 e^2 n_0}}.$$

- B5 Find the distance λ (formula and numerical value) at which the potential $\varphi(z)$ decreases by a factor of e . This length is considered the characteristic size of the Gouy diffuse layer. Liquid viscosity $\eta = 8.9 \cdot 10^{-4}$ Па · с, ion concentration $n = 1.1 \cdot 10^{25}$ м⁻³, electrical constant $\varepsilon_0 = 8.85 \cdot 10^{-12}$ Ф/м, Boltzmann constant $k = 1.38 \cdot 10^{-23}$ Дж/К, temperature $T = 300$ К, dielectric constant $\varepsilon = 100$. 0.50**

The potential decreases by a factor of e at $z = \lambda$, the formula for λ was found in paragraph B4.

Answer: $\lambda = \sqrt{\frac{\varepsilon \varepsilon_0 k T}{2 e^2 n_0}} \approx 2,55 \text{ нм}$

- C1 Write down Newton's second law for a small volume of liquid in projection onto the direction of the electric field, taking into account viscous friction and the electric field acting on the uncompensated charge. Express the answer in terms of the electric field E , the ion concentration n_0 , the potential $\varphi(z)$, the fluid and capillary parameters, and the velocity $v(z)$ and its derivatives. 0.80**

This point is similar to the solution to point A1. The viscous friction forces acting on the layer will be equivalent, and the electric field will act with a force of

$$F_E = \rho E a L dz$$

Тогда аналогично уравнению (1):

$$\eta \frac{d^2 v}{dz^2} + \rho E = 0$$

С учетом выражений (5) и (7):

$$\eta \frac{d^2 v}{dz^2} - \frac{\varepsilon \varepsilon_0 E}{\lambda^2} \varphi_\delta \exp(-z/\lambda) = 0$$

Answer:

$$\eta \frac{d^2 v}{dz^2} - \frac{\varepsilon \varepsilon_0 E}{\lambda^2} \varphi_\delta \exp(-z/\lambda) = 0$$

- C2 Find the dependence of the fluid flow velocity on the coordinate $v(z)$. 1.20**

From point C1 we have the differential equation:

$$\frac{d^2 v}{dz^2} = \frac{\varepsilon \varepsilon_0 E}{\eta \lambda^2} \varphi_\delta e^{(-z/\lambda)}$$

Найдем его решение, проинтегрировав обе части по dz дважды:

$$v = \frac{\varepsilon\varepsilon_0 E}{\eta} \varphi_\delta e^{(-z/\lambda)} + C_1 z + C_2 \quad (C_1, C_2 = \text{const})$$

Из граничного условия $v(0) = 0$ находим:

$$C_2 = -\frac{\varepsilon\varepsilon_0 E}{\eta} \varphi_\delta$$

Со вторым граничным условием всё обстоит немного сложнее. Строго говоря, *correct* граничным условием является $v(b) = 0$. Однако для того, чтобы рассматривать вторую стенку необходимо внести изменения и в распределение потенциала, добавив возрастающую экспоненту. Но с точки зрения физики это только добавит такую же картинку распределения потенциала у другой стенки. При рассмотрении же только одной стенки возникает следующая проблема: с используемой точностью при $z \sim \lambda$ любые граничные условия для $z \sim b$ будут неотличимы друг от друга, так как фактически они все эквивалентны бесконечности и члены вида $z/b \ll 1$ пренебрежимо малы. Тем не менее если мы хотим найти распределение потенциала на масштабах размеров трубы, вклад этих членов становится существенным. В нашей же модели найти C_2 можно следующим образом. Обратимся к формуле (8) – она справедлива при любом $\varphi(z)$. Т.к. граничные условия на стенках одинаковы, $v(z)$ и $\varphi(z)$ должны быть симметричны относительно центра трубы. Из этого факта и формулы (8) следует $C_1 = 0$.

Примечание. In fact, it is possible to obtain a more general expression for the speed v , showing that it depends only on $\varphi(z)$, while the form of the dependence of the potential on the coordinate itself is unimportant. Let's write the equation:

$$\eta \frac{d^2 v}{dz^2} + \rho E = 0$$

С учетом теоремы Гаусса оно запишется в виде:

$$\eta \frac{d^2 v}{dz^2} - \varepsilon\varepsilon_0 \frac{d^2 \varphi}{dz^2} E = 0$$

Аналогично два раза проинтегрируем по dz и получим:

$$v = \frac{\varepsilon\varepsilon_0 E \varphi}{\eta} + C_3 z + C_4 \quad (C_3, C_4 = \text{const})$$

$C_3 = 0$ в силу симметрии трубки. При $\varphi = \varphi_\delta$ $v = 0$, откуда $C_4 = -\frac{\varepsilon\varepsilon_0 E \varphi_\delta}{\eta}$. Итоговый ответ:

$$v = \frac{\varepsilon\varepsilon_0 E}{\eta} (\varphi - \varphi_\delta) \quad (8)$$

Answer:

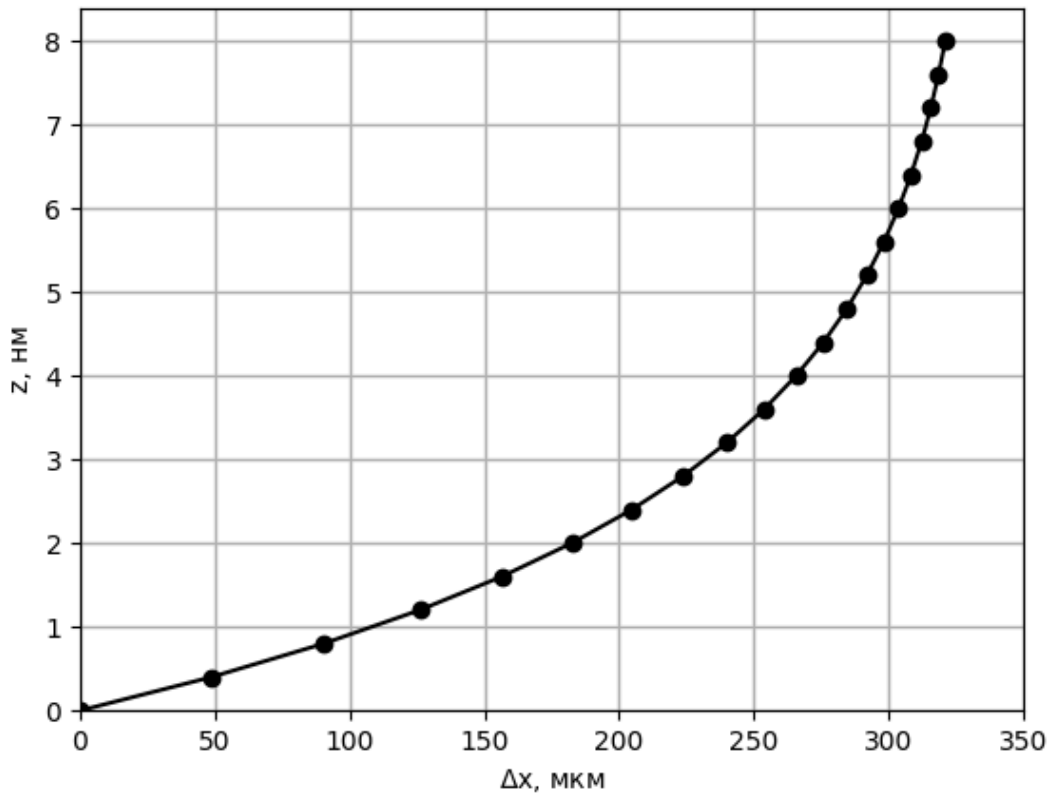
$$v = -\frac{\varepsilon\varepsilon_0 E}{\eta} \varphi_\delta \left(1 - e^{-z/\lambda}\right)$$

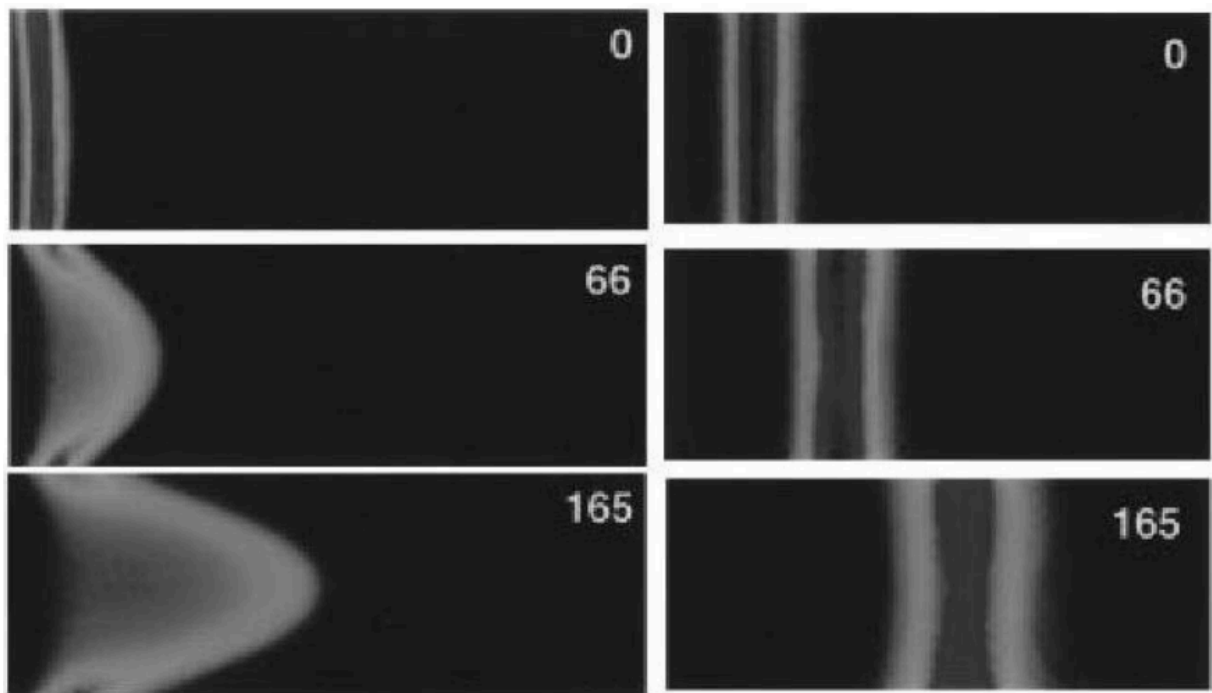
part A. On your answer sheet, draw where the reactant will be located through $\tau = 5 c$. Use the capillary parameters from A2 and the numerical data from B5. Potential $\varphi_\delta = 1.8 B$, voltage drop along the length of the capillary $U = 0.3 B$. Neglect the diffusion of the reagent outside the diffusion layer. For the drawing, select an appropriate scale along the horizontal axis directed along the direction of fluid flow.

Using the formula for the speed obtained at point C2, the displacement of the liquid layer ($\Delta x = v\tau$) is easily calculated. Once you have the values for several z , you can plot the graph shown below.

Note that the exponential decays significantly already at $z \sim \lambda$. Considering that $\lambda \ll b$, with $z \sim b$ $v \approx -\frac{\varepsilon\varepsilon_0 E}{\eta}\varphi_\delta = \text{const.}$, and in the figure the front will be depicted as a straight line parallel to the z axis.

Photos of the real experiment. On the left is the reagent distribution profile in the case of uniformly distributed pressure, on the right is the case considered in part C.





C4 Based on part C of this problem, propose a system that allows mixing layers of liquid flowing near the wall of a capillary. 1.00

Answer: In this part, various systems are possible; we will give a qualitative example of one of them. Let us denote by Ox the axis codirectional with $\vec{E}$ in the figure from part C. Let the projection of the field onto this axis have the following form depending on the coordinate x :

$$E_x = E_0 \sin(kx)$$

На основании предыдущих пунктов части C we can argue that this will lead to the fact that the layers in some places will be forced to move towards each other. This movement will cause them to mix.